

2(a)(i)	<i>independent variable:</i> type / AW , of trypsin ; <i>dependent variable:</i> time (taken to break down gelatine) ;	2
2(a)(ii)	any one from: 1 pH (of solution) ; 2 surface area / mass / volume, of cube ; 3 concentration of, gelatine / collagen, in cube ;	1
2(b)(i)	<i>mean</i> = 131.1 ;	1
2(b)(ii)	<i>s</i> = 3.2 ;	1
2(b)(iii)	any three from: 1 critical value (at degrees of freedom = 22 and $p = 0.05$) = 2.074 ; 2 calculated (t -) value / 2.663 is, greater than, critical (t -) value / 2.074 ; ora 3 null hypothesis is rejected ; 4 there is a <u>significant difference</u> (at $p = 0.05$) ;	3

Question	Answer	Marks
2(b)(iv)	any two from: 1 <i>idea that</i> salmon trypsin is adapted to colder temperatures ; ora 2 (at 20 °C) salmon trypsin forms enzyme–substrate complexes at a faster rate (than pig trypsin) ; ora 3 AVP ;	2